

Instances, images and volumes



Day 2 - Morning

Running scenario: Thara Hospitals patient portal pilot

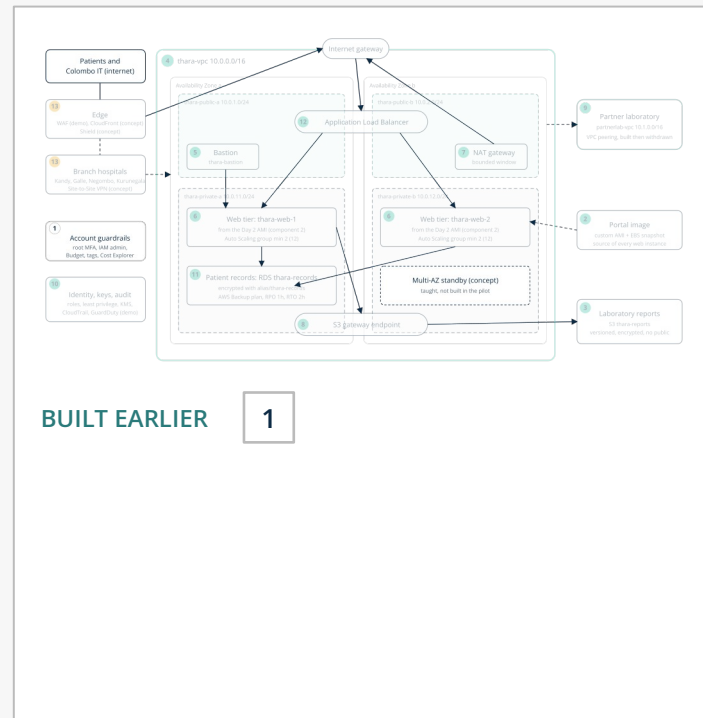
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Today

The account is locked. Today the first server and the first bucket go in.

- The EC2 model - 35 min
- Block storage and images - 30 min
- Object storage, first look - 20 min
- Paying for compute - 25 min



CONCEPT 1



The EC2 model

The portal web server is the first thing Thara will run in the cloud.

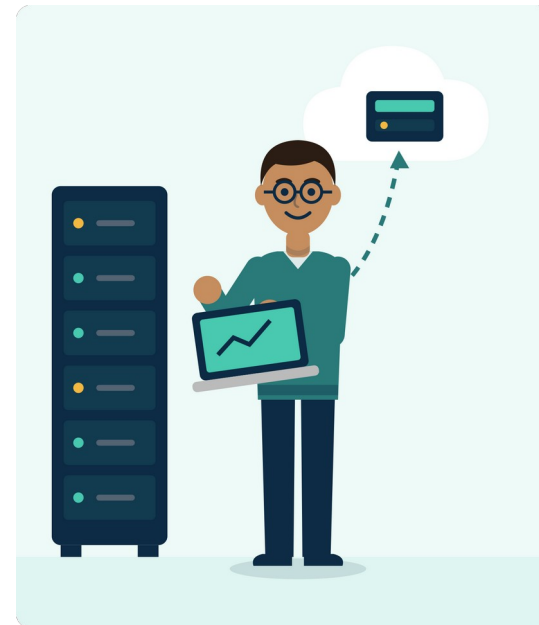
35 min

What an instance type tells you

- `t3.micro` reads as family, generation, size
- The family sets the balance of CPU, memory and storage; the size sets how much
- **t3.micro**: 2 vCPUs, 1 GiB, burstable. The module's workhorse

YOU ALREADY KNOW THIS

A rack server built from a golden image, with a KVM console for when the network is down.



One small server, launched from a laptop.

Four things come together at launch

- **AMI**: the image the disk is built from. The unit of reproducibility
- **Key pair**: AWS keeps the public half; you download the private half once
- **User data**: a script that runs once, at first boot, as root
- **Instance metadata**: the instance asks about itself at 169.254.169.254



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User data installs the web server and writes one page: Thara Hospitals portal (pilot).

Three ways to get a shell

Instance Connect

60 s

In the browser

AWS pushes a temporary key to the instance for one minute.

Used in today's lab.

Session Manager

No port

Through an agent

No SSH port is open at all. Access is controlled by IAM.

Named, not used.

SSH

Your key

From your laptop

Your private key file opens the door for as long as it exists.

Used on Day 3.

A private key file never expires. Whoever holds a copy can log in.

A firewall in front of every instance



- A security group is a firewall attached to the instance
- It holds allow rules only; anything not allowed is dropped
- It is **stateful**: a reply to allowed traffic is allowed automatically

Today you write three rules. Day 3 treats security groups in depth.

CONCEPT 2

Block storage and images

The portal image must be rebuildable after the next flood.

30 min

A volume is a disk that lives in one zone

- EBS is block storage: attach it, format it, mount it
- **gp3** is the default type, and the only one this module needs
- A volume attaches only to an instance in its own availability zone

YOU ALREADY KNOW THIS

A SAN LUN versus a local scratch disk. The LUN outlives the server.

COMMON MISTAKE

Stopping an instance stops all billing.

The instance charge stops. Its volumes and snapshots still bill.

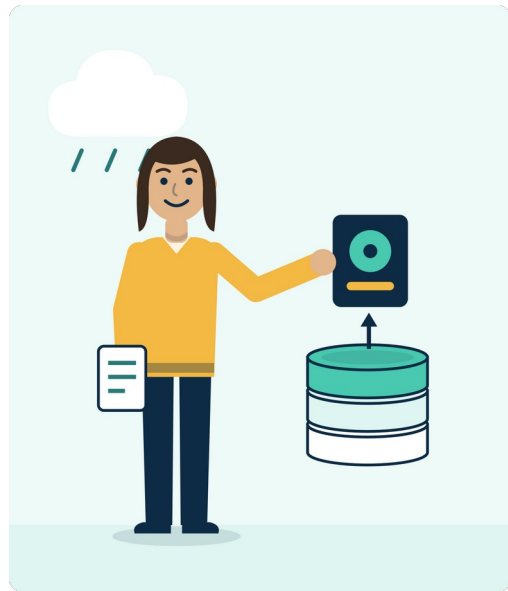
An image is a snapshot plus launch metadata

- A snapshot is a point-in-time copy, kept in S3-backed storage
- Snapshots are **incremental**: later ones store changed blocks only
- An AMI points at a snapshot and adds what a launch needs

COMMON MISTAKE

A snapshot is a full copy every time.

Only the first one is. Any one of them still restores the whole volume.



The snapshot holds the bytes; the AMI holds the instructions.

Where does the data live when the instance stops?

On an EBS volume

The volume is separate from the host. Stop, start or reboot the instance and the data is still there.

Snapshot it, and it can be rebuilt in another zone.

On instance store

The disk is inside the host machine. Stop the instance, or lose the host, and the data is gone for good.

Outside this module, except as this warning.

Thara lost its servers and the only record of how they were built. An image is the record.

CONCEPT 3

Object storage, first look

Laboratory report PDFs, and the story of the leaked link.

20 min

Objects and keys, not files and folders

- S3 stores whole objects under a key; the key is one long name
- You put, get or delete an object. You do not edit the middle of it
- Bucket names are global; the bucket and its data live in one region

YOU ALREADY KNOW THIS

A document management system versus a file share.

COMMON MISTAKE

S3 is a file system.

It is a key-value object store with no real directories.

Storage classes, as vocabulary

Standard

Often

The default

For data that is read regularly.

New reports.

Infrequent Access

Rarely

Cheaper to store

A lower storage price, and a fee each time you read.

Older reports.

Glacier classes

Archive

Cheapest to store

Reading back takes from milliseconds to hours, by class.

Reports kept for years.

Day 6 moves reports between classes automatically.

Block Public Access is on, and stays on

- Every new bucket blocks public access by default
- It is a master switch: it overrides any setting that would expose an object

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A patient's report link opened for anyone who had it. With this switch on, a report cannot be made public by accident.



Reports go in as objects. The bucket stays private.

CONCEPT 4

Paying for compute

Why would a one-year commitment cut the web tier's cost?

25 min



CFO

CHIEF FINANCIAL OFFICER

You tell me a one-year commitment would cut the web tier's cost by roughly a third. Why, and what am I promising in return?

The Chief Financial Officer, on paying for compute.

Four ways to pay for the same server

0.011

USD per hour on demand
t3.micro, at the time of writing

1/3

Off for one year
roughly

2 min

Spot notice
before AWS takes it back

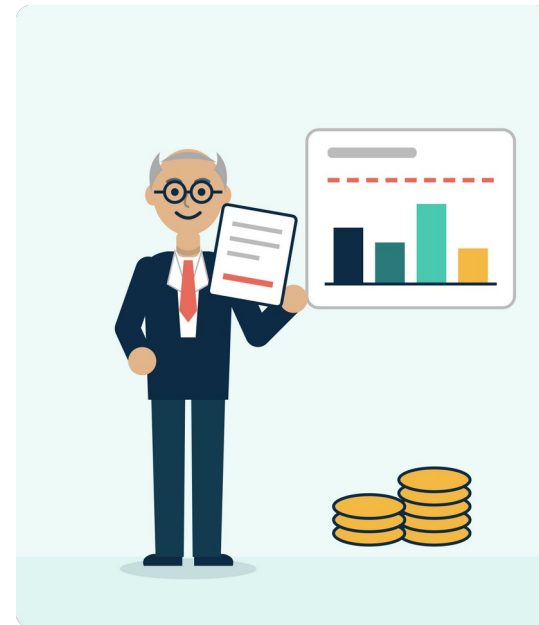
Option	What you give up	Discount
On-demand	Nothing. Stop whenever you like	None
Spot	Certainty. AWS can reclaim the instance	Often over half
Reserved Instances	Flexibility. One or three years, one family	About a third
Savings Plans	Flexibility. A set spend per hour	Similar

The Free Plan blocks commitments

- A Free Plan account cannot buy Reserved Instances or Savings Plans
- You must still explain what each option trades for its discount

YOU ALREADY KNOW THIS

A leased line contract versus burstable billing.



A commitment is a promise to pay, used or not.

Tags from today, and two names for later

- Every resource from today: `Module=CC, Day=2, Owner=<student id>`
- On Day 7 you group your spending by these tags in Cost Explorer
- **ECS** runs containers for you across a group of machines
- **Lambda** runs one function per event, with no server to manage

ECS and Lambda: know what they are for. You will not build with them.

Demonstration: Launch, connect, image



1. Launch a `t3.micro` from the Amazon Linux AMI, with user data that serves the pilot page
2. Connect with Instance Connect; ask the metadata service who we are; show the tags
3. Stop and start: the public IP has changed. Elastic IPs wait for Day 3
4. Create an AMI from the instance and find it in the AMI list

Watch closely: this is the first half of your lab.

Common mistakes



COMMON MISTAKE

Stopping an instance stops all billing.

EBS volumes and snapshots still bill.

COMMON MISTAKE

A snapshot is a full copy every time.

It is incremental after the first.

COMMON MISTAKE

S3 is a file system.

It is a key-value object store with no real directories.

Before the next session



READ

Wittig and Wittig

The chapters on virtual machines (EC2) and on block storage (EBS).

SELF-CHECK

CLF-C02 bank

Domain 3 compute set, 15 items.

PRACTICE

PQ2

Storage for the reports and the records, with cost reasoning.

Evidence Pack 2 follows this afternoon's lab.

Questions & Discussion



Next: this afternoon you build components 2 and 3 yourself.



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